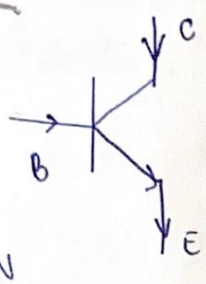
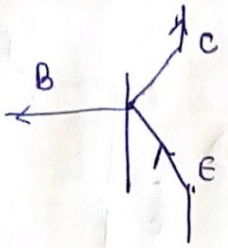
- Seminar 24 Noiembrie 2022 -



NPN

PNP



R.A.N

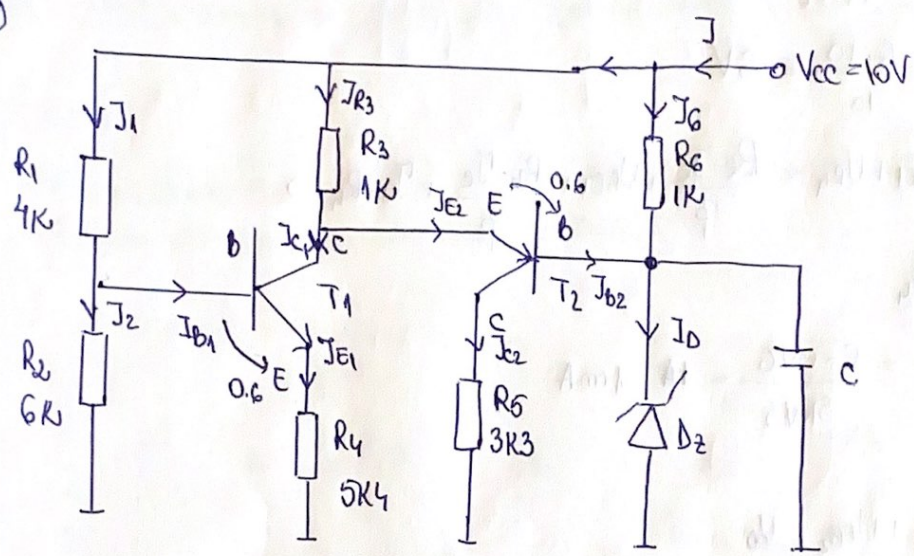
$I_C = \beta I_B$

$U_{BE} = 0.6$

$U_{EB} = 0.6$

P.S.F.

①



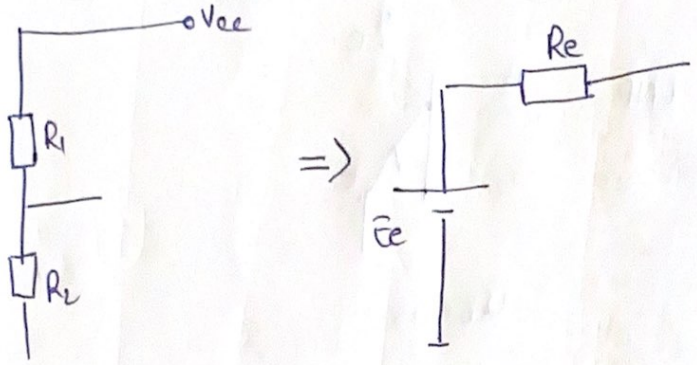
$U_{Dz} = 7V4$ $\beta = 100$

PSF = T_1, T_2 (I_C, U_{CE})

→ sensul curentului prin diodă: de la tensiune mare la tensiune mică.

Pp $I_D > 0 \Rightarrow U_{Dz} = 7V4$, deoarece $U_D = U_{Dz}$ din cauza polarizării inverse

Facem sistemul echivalent =>



$Re = R_1 \| R_2 \Rightarrow \frac{R_1 R_2}{R_1+R_2} = \frac{24}{10} = 2{,}4\,K = R_{polarizat}$

$Ee = \frac{R_2 \cdot E}{R_1+R_2} = \frac{6 \cdot 10}{10} = 6V$

$Ee = Re \cdot I_{B1} + U_{BE1} + U_{R4} = \frac{Re \cdot I_C}{\beta} + U_{BE1} + \frac{R_4 \cdot I_C}{\beta} + I_C \cdot R_4 \Rightarrow$

$\Rightarrow Ee = U_{BE1} + I_C R_4$

$\Rightarrow I_C = \frac{Ee - U_{BE1}}{R_4} = \frac{6 - 0V6}{5K\Omega} = 1mA$

$V_{cc} - 0 = I_{R_3} R_3 + U_{EB_2} + U_D$

$\Rightarrow I_{R_3} = \frac{V_{cc} - U_{EB_2} - U_D}{R_3}$

$I_{R_3} = 2mA$

$\Rightarrow I_{R_3} = I_{C1} + I_{E2} \Rightarrow I_{E2} \sim I_{C2} \Rightarrow I_{C2} = I_{R_3} - I_{C1} = 2mA - 1mA = 1mA$

$V_{cc} - 0 = U_{R_3} + U_{R_4} + U_{CE1} \Rightarrow U_{CE1} = 2V6 \Rightarrow V_{CE1} > U_{BE} \Rightarrow (A) \Rightarrow$ tr. e în RAN

$V_{cc} - 0 = I_{E_3} R_3 + U_{EC_2} + I_{C_2} R_5 \Rightarrow U_{EC_2} = 10V - 2 - 1 \cdot 3K3 =$

$= 10 - 2 - 3.3 =$

$= 10 - 5.3 = 4V7 > 0.6 \Rightarrow (A)$

$\Rightarrow$ tr. e în RAN

~~$U_{BE2} + U_{R5} = 0$~~

~~$U_{D2} = U_{BE2}$~~

$V_{cc} - 0 = I_6 R_6 + U_{D2}$

$I_6 R_6 = V_{cc} - U_{D7}$

$I_6 = \frac{V_{cc} - U_{D2}}{R_6}$

$I_6 = \frac{10 - 7.4}{1} = 2.6\ mA$

$I_D = I_{BE2} + I_6$

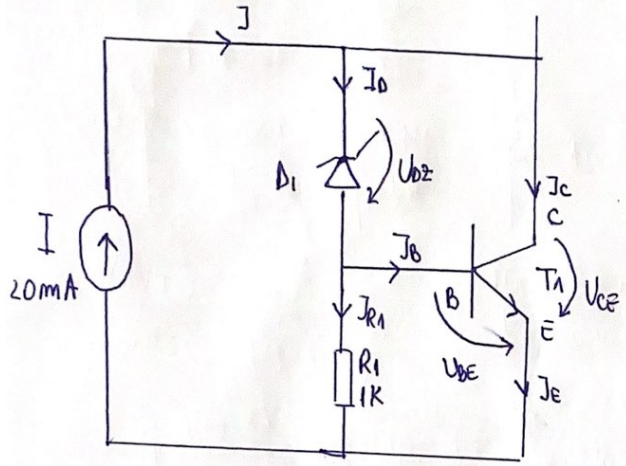
$I_D = \frac{I_{c2}}{\beta} + I_6$

$I_D = \frac{1\ mA}{100} + 2\ mA\ 6 \cong 2.6\ mA > 0 \Rightarrow$ presupunerea este corectă

CASCODA

→ Putem elimina condensatorul a.î. nu se schimbă nimic în PSF

→ Putem elimina R_5 ⇒ nu schimbă curenții



$\beta = 100$

$U_{DZ} = 4V4$

P.S.F.

$I_{R1} \cdot R_1 = U_{BE} \Rightarrow I_{R1} = 0.6\ mA$

$I_D = I_{R1} + I_B$

$I = I_D + I_C$

$I_C = I - I_D$

$I_D = I_{R1} + \frac{I_C}{\beta}$ ⇒

$I_C + \frac{I_C}{\beta} = I_{R1}$

$I_C = I - I_D$

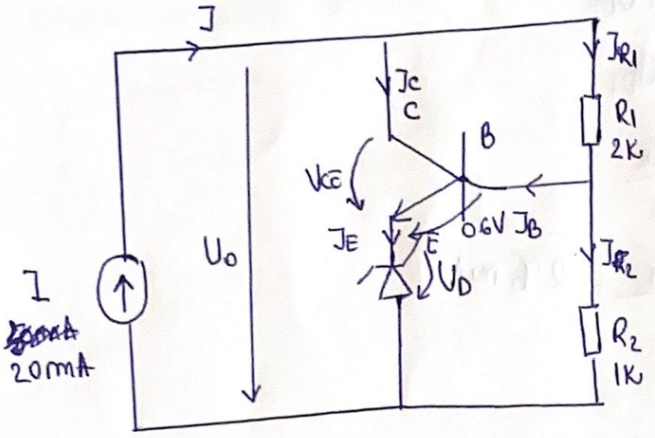
$I_C = \frac{I - I_{R1}}{1 + \frac{1}{\beta}} = 20 - 0,6 = 19,4\ mA$

~~U_{DZ}~~

$U_{DZ} + U_{BE} = U_{CE}$

$U_{CE} = 4V4 + 0V6 = 5V$

→ Dioda Zenner, tine tensiunea constantă indiferent de curentul care trece prin ea.



$\beta = 100$

$U_{DZ} = 4V4$

PSF, $U_O = ?$

$I = J_C + J_D$

$J_1 = J_B + J_2$

$J_C = J_B + J_E$

$U_{BE} + U_D = U_{R2} \Rightarrow$

$U_{R2} = 0.6 + 4.4 = 5V$

$\Rightarrow J_2 = \frac{U_{R2}}{R_2} = \frac{5V}{1K} = 5mA$

$\Rightarrow J_1 = \frac{J_C}{\beta} + J_2 \Rightarrow J_C = J - J_2 - \frac{J_C}{\beta} \Rightarrow J_C \left(1 + \frac{1}{\beta}\right) = J - J_2$

$J_C = \frac{J - J_2}{1 + \frac{1}{\beta}} = 20mA - 5mA = 15mA$

$U_{CE} = U_{R1} + U_{BE} = 10.6V$

$U_O = V_{CE} + U_D = 10.6V + 4.4V = 15V.$